

Question Shuffling Allowed :

No

Question Id : 6406531415432 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (114 to 115)

Question Label : Comprehension

$$\text{Let } A = \begin{bmatrix} i & 0 & 0 \\ -i & i & 0 \\ 1 & 0 & -i \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 114 Question Id : 6406531415433 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Suppose λ_1, λ_2 and λ_3 are the eigenvalues of $2A^2 - 3I$, find $\lambda_1\lambda_2\lambda_3$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-125

Question Number : 115 Question Id : 6406531415434 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is /are true?

Options :

6406534736687. ✓ A is not a Hermitian matrix.

6406534736688. ✓ A is not a unitary matrix.

6406534736689. ✗ A is diagonalizable.

6406534736690. ✗ None of these

Java

Section Id :

64065399547

Section Number :

6

Section type :

Online

Mandatory or Optional :	Mandatory
Number of Questions :	24
Number of Questions to be attempted :	24
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653228427
Question Shuffling Allowed :	No

Question Number : 116 Question Id : 6406531415435 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : PROGRAMMING CONCEPTS USING JAVA (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406534736691.  YES

6406534736692.  NO

Sub-Section Number :	2
Sub-Section Id :	640653228428
Question Shuffling Allowed :	Yes

Question Number : 117 Question Id : 6406531415436 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java program. How many activation records exist simultaneously on the call stack when the call stack reaches its maximum depth, during the execution of the program?

```
class Calculator {
    public static int compute(int n) {
        if (n <= 1) {
            return 1;
        }
        else {
            return compute(n - 1) + helper(n);
        }
    }
    public static int helper(int x) {
        return x * 2;
    }
}

class TestCallStack {
    public static void main(String[] args) {
        int result = Calculator.compute(4);
        System.out.println(result);
    }
}
```

Options :

6406534736693. ✖ 4

6406534736694. ✔ 5

6406534736695. ✖ 6

6406534736696. ✖ 8

Question Number : 118 Question Id : 6406531415437 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the code given below that checks whether two customers are the same. Method `equals` is overridden to compare two `Customer` objects as follows.

If two customers have the same name and `customerId`, then they are considered the same.

```
class Customer {
    private String name;
    private int customerId;
    // Constructor to initialize instance variables
    public boolean equals(Object obj) {
        //CODE BLOCK
    }
}

public class TestCustomer {
    public static void main(String[] args) {
        Customer c1 = new Customer("Keerthi", 101);
        Customer c2 = new Customer("Keerthi", 102);
        if (c1.equals(c2))
            System.out.println("c1 and c2 are same");
        else
            System.out.println("c1 and c2 are different");
    }
}
```

Choose the correct option to fill in place of CODE BLOCK so that the output is:
c1 and c2 are different

Options :

```
if(obj instanceof Customer) {
    if(this.name == obj.name && this.customerId == obj.customerId)
        return true;
}
```

6406534736697. ✖ return false;

```
if(this.name == obj.name && this.customerId == obj.customerId)
    return true;
```

6406534736698. ✖ return false;

```
if(obj instanceof Customer) {
    Customer c = obj;
    if(this.name == c.name && this.customerId == c.customerId)
        return true;
}
```

6406534736699. ✖ return false;

6406534736700. ✔

```

if(obj instanceof Customer) {
    Customer c = (Customer) obj;
    if(this.name == c.name && this.customerId == c.customerId)
        return true;
}
return false;

```

Question Number : 119 Question Id : 6406531415438 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```

interface FuelDispenser {
    public default void dispenseFuel() {
        System.out.println("Dispensing regular petrol");
    }
}

interface BillingSystem {
    public default void generateInvoice() {
        System.out.println("Generating standard invoice");
    }
}

class PetrolPump implements FuelDispenser, BillingSystem {
    public void dispenseFuel() {
        System.out.println("Dispensing petrol with additives");
    }
}

public class TestPetrolPump {
    public static void main(String[] args) {
        FuelDispenser fd = new PetrolPump();
        fd.dispenseFuel();
        fd.generateInvoice(); // LINE 1
    }
}

```

Choose the correct option.

Options :

This program generates the output:

Dispensing premium petrol with additives

6406534736701. ✖ Generating standard invoice

6406534736702. ✖

This program generates the output:

Dispensing regular petrol

Generating standard invoice

6406534736703. ✓ LINE 1 generates compiler error because fd of type FuelDispenser cannot invoke method generateInvoice().

6406534736704. ✖ This program generates compiler error because neither is class PetrolPump declared as abstract nor does it override method generateInvoice().

Question Number : 120 Question Id : 6406531415439 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.util.*;
class Investor {
    String name, company;
    double investment;
    public Investor(String n, String c, double i){
        name = n;
        company = c;
        investment = i;
    }
}

public class ShareMarket {
    public static void calculateInvestments(ArrayList<Investor> invList) {
        var map = new TreeMap<String, Double>();
        for (Investor inv : invList) {
            map.put(inv.name, map.getDefault(inv.name, 0.0) + inv.investment);
        }
        for (Map.Entry<String, Double> e : map.entrySet()) {
            System.out.println(e.getKey() + " = " + e.getValue());
        }
    }

    public static void main(String[] args) {
        ArrayList<Investor> invList = new ArrayList<Investor>();
        invList.add(new Investor("Ravi", "Infosys", 150000));
        invList.add(new Investor("Sneha", "TCS", 200000));
        invList.add(new Investor("Ravi", "Wipro", 100000));
        invList.add(new Investor("Anil", "Infosys", 180000));
        calculateInvestments(invList);
    }
}
```

What will the output be?

Options :

Anil = 180000.0

Ravi = 150000.0

6406534736705. ✖ Sneha = 200000.0

Anil = 180000.0

Ravi = 250000.0

6406534736706. ✔ Sneha = 200000.0

Anil = 180000.0

Ravi = 150000.0

Ravi = 100000.0

6406534736707. ✖ Sneha = 200000.0

Anil = 180000.0
Ravi = 100000.0
6406534736708. ✖ Sneha = 200000.0

Question Number : 121 Question Id : 6406531415440 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java code that uses chained exceptions.

```
class TransmissionException extends Exception {
    public TransmissionException(String msg) {
        super(msg);
    }
}

class PacketException extends Exception {
    public PacketException(String msg) {
        super(msg);
    }
}

public class NetworkTest {
    public static void sendPacket() throws TransmissionException {
        try {
            preparePacket();
        } catch (PacketException e) {
            TransmissionException te =
                new TransmissionException("Transmission failed");
            te.initCause(e);
            throw te;
        }
    }

    public static void preparePacket() throws PacketException {
        throw new PacketException("Packet corrupted");
    }

    public static void main(String[] args) {
        try {
            sendPacket();
        } catch (TransmissionException e) {
            System.out.println(e.getMessage());
            System.out.println(e.getCause().getMessage());
        }
    }
}
```

Choose the correct option.

Options :

This program generates the output:

6406534736709. ✖ Transmission failed

This program generates the output:

6406534736710. ✖ Packet corrupted

This program generates the output:

Transmission failed

6406534736711. ✔ Packet corrupted

This program generates the output:

Packet corrupted

6406534736712. ✖ Transmission failed

Question Number : 122 Question Id : 6406531415441 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java code.

```
class BankAccount {
    private String accountHolder;
    private double balance;
    public BankAccount(String ac, double in) {
        assert ac != null : "Account holder cannot be null"; // assert-1
        assert ac != "" : "Account holder cannot be empty"; // assert-2
        assert in >= 0 : "Initial balance cannot be negative"; // assert-3
        accountHolder = ac;
        balance = in;
    }
    public void transferTo(BankAccount target, double amount) {
        assert target != null : "Target account cannot be null"; // assert-4
        if (amount > 0 && balance >= amount) {
            balance -= amount;
            target.balance += amount;
        }
    }
    public double getBalance() {
        return balance;
    }
}

public class TransferTest {
    public static void main(String[] args) {
        BankAccount sender = new BankAccount("", 1000.0);
        BankAccount receiver = new BankAccount("Alex", 500.0);
        sender.transferTo(receiver, 200.0);
    }
}
```

Identify the first assert statement that throws the AssertionError when the class is executed as: `java -ea TransferTest`

Options :

6406534736713. ✖ assert-1

6406534736714. ✔ assert-2

6406534736715. ✖ assert-3

6406534736716. ✖ assert-4

Question Number : 123 Question Id : 6406531415442 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following code and answer the question that follows.

```
import java.util.*;
import java.util.stream.*;
public class MapTest {
    public static void main(String[] args) {
        List<Integer> numbers = Arrays.asList(4, 7, 9, 12, 14, 15, 18);

        numbers.stream()
            .filter(n -> n % 3 == 0)
            .map(n -> n * 2)
            .skip(1)
            .limit(2)
            .forEach(n -> System.out.print(n + " "));
    }
}
```

What will the output be?

Options :

6406534736717. ✖ 18 24

6406534736718. ✖ 20 28

6406534736719. ✔ 24 30

6406534736720. ✖ 18 36

Question Number : 124 Question Id : 6406531415443 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Assume that the file `animals.txt` contains the following lines of text in it.

Lions live in the savannah.
Elephants are the largest land animals.
Penguins cannot fly.

Consider the code given below.

```
import java.io.*;
import java.util.*;
public class AnimalFacts {
    public static void main(String[] args) {
        try {
            var input = new FileInputStream("animals.txt");
            var sc = new Scanner(input);          // LINE 1
            System.out.println("Animal facts:");
            System.out.println(sc.nextLine());
            System.out.println(sc.next());
            System.out.println(sc.nextLine());
        }
        catch (FileNotFoundException e) {
            System.out.println("File not found.");
        }
        catch (IOException e) {
            System.out.println("An I/O error occurred.");
        }
    }
}
```

Choose the correct option.

Options :

6406534736721. ✖ LINE 1 throws an IOException.

This program generates the output:

Animal facts:

Lions live in the savannah.

Elephants are the largest land animals.

6406534736722. ✖ Penguins cannot fly.

This program generates the output:

Animal facts:

Lions live in the savannah.

Elephants

6406534736723. ✔ are the largest land animals.

This program generates the output:

Animal facts:

Lions live in the savannah.

Lions

6406534736724. ✖ Penguins cannot fly.

Question Number : 125 Question Id : 6406531415444 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.io.*;
class Employee implements Serializable {
    private String name = "Unknown";
    private transient double salary = 50000.0;
    private String department = "General";
    public Employee(String n, double s, String d) {
        name = n;
        salary = s;
        department = d;
    }
    public String toString() {
        return name + ", " + salary + ", " + department;
    }
}

public class Test {
    public static void main(String[] args) throws Exception {
        var fos = new FileOutputStream("employee.txt");
        var os = new ObjectOutputStream(fos);
        os.writeObject(new Employee("James", 70000.0, "IT"));
        var fis = new FileInputStream("employee.txt");
        var ois = new ObjectInputStream(fis);
        Employee emp = (Employee) ois.readObject();
        System.out.println(emp);
    }
}
```

What will the output be?

Options :

6406534736725. ✖ Unknown, 50000.0, General

6406534736726. ✖ James, 70000.0, IT

6406534736727. ✔ James, 0.0, IT

6406534736728. ✖ James, 70000.0, General

Question Number : 126 Question Id : 6406531415446 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the code given below.

```
class NumberPrinter extends Thread {
    private int number;
    public NumberPrinter(int n) {
        number = n;
    }
    public void run() {
        for (int i = 0; i < 4; i++) {
            System.out.print(number + " ");
            try {
                sleep(200);
            } catch (InterruptedException e) {
                e.printStackTrace();
            }
        }
    }
}

public class SequenceTest {
    public static void main(String[] args) throws InterruptedException {
        Thread t1 = new NumberPrinter(10);
        Thread t2 = new NumberPrinter(20);
        Thread t3 = new NumberPrinter(30);
        t1.start();
        t1.join();
        t2.start();
        t3.start();
    }
}
```

Choose the correct option.

Options :

6406534736733. ✖ It prints 10 four times, 20 four times, and 30 four times in an interleaved manner.

6406534736734. ✔ It prints 10 four times, followed by 20 and 30 (four times each) in an interleaved manner.

6406534736735. ✖ It prints 10 and 20 (four times each) in an interleaved manner, followed by 30

four times.

6406534736736. ✖ It prints 10 four times, followed by 20 four times, followed by 30 four times.

Question Number : 127 Question Id : 6406531415447 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.util.*;
class MarksUpdater extends Thread {
    private Map<String, Integer> studentMarks;
    public MarksUpdater(Map<String, Integer> marks) {
        studentMarks = marks;
    }
    public void run() {
        studentMarks.put("Priya", 91);
    }
}
public class StudentMarksList {
    public static void main(String[] args) throws InterruptedException {
        Map<String, Integer> marks = new LinkedHashMap<>();
        marks.put("Amit", 88);
        marks.put("Sneha", 93);
        marks.put("Ravi", 76);
        Thread updater = new MarksUpdater(marks);
        updater.start();
        updater.join();
        for (Map.Entry<String, Integer> entry : marks.entrySet()) {
            System.out.println(entry.getKey() + " => " + entry.getValue());
        }
    }
}
```

Which of the following is true about the given code?

Options :

6406534736737. ✖ The program may throw a ConcurrentModificationException.

This program generates the output:

Amit => 88

Sneha => 93

Ravi => 76

6406534736738. ✔ Priya => 91

6406534736739. ✖

This program generates the output:

Sneha => 93

Priya => 91

Ravi => 76

Amit => 88

This program may generate the output:

Amit => 88

Sneha => 93

6406534736740. ✖ Ravi => 76

Question Number : 128 Question Id : 6406531415449 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
class Actor implements Cloneable {
    private String name;
    private String[] movies;

    public Actor(String n, String[] m) {
        name = n;
        movies = m;
    }
    // method to set name
    // method to update movie at a particular index
    // method to get name
    // method to get movie at index

    public Object clone() throws CloneNotSupportedException {
        return super.clone();
    }
}

public class MovieTest {
    public static void main(String[] args) throws CloneNotSupportedException {
        String[] m = {"Action Hero", "Comedy Nights", "Romantic Saga"};
        Actor a1 = new Actor("Raj", m);
        Actor a2 = (Actor) a1.clone();
        Actor a3 = a1;

        a2.setMovie(0, "Thriller Hit");
        a3.setName("Arjun");

        System.out.println(a1.getName() + " " + a1.getMovie(0));
        System.out.println(a2.getName() + " " + a2.getMovie(0));
        System.out.println(a3.getName() + " " + a3.getMovie(0));
    }
}
```

Choose the correct option.

Options :

Arjun Action Hero

Raj Thriller Hit

6406534736745. ✖ Arjun Action Hero

Raj Action Hero

Raj Thriller Hit

6406534736746. ✖ Arjun Action Hero

6406534736747. ✔

Arjun Thriller Hit
Raj Thriller Hit
Arjun Thriller Hit

Raj Thriller Hit
Raj Thriller Hit
Arjun Thriller Hit

6406534736748. ✖

Question Number : 129 Question Id : 6406531415450 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Method `Optional.ofNullable(T value)` returns an `Optional` that describes the specific value, if non-null; otherwise returns an `Optional.empty`.

Based on this description, consider the code given below and answer the question that follows.

```
import java.util.*;
class StateInfo {
    public String getFamousLake(String state) {
        String result;
        switch(state) {
            case "Jammu and Kashmir": result = "Dal Lake";
                                    break;
            case "Madhya Pradesh": result = "Upper Lake";
                                    break;
            default: result = null;
                    break;
        }
        return result;
    }
}

public class OptionalTest {
    public static void main(String[] args){
        Optional<String> op = Optional.ofNullable(new StateInfo()
                                           .getFamousLake("Kerala"));
        op.ifPresent(lake -> System.out.println(lake.toUpperCase()));
    }
}
```

Choose the correct option.

Options :

6406534736749. ✔ This program does not generate any output.

6406534736750. ✖ This program generates a NullPointerException.

6406534736751. ✖ This program generates the output: NULL

6406534736752. ✖ This program generates the output: DAL LAKE

6406534736753. ✖ This program generates the output: UPPER LAKE

Question Number : 130 Question Id : 6406531415451 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the following Java code.

```
class Employee {
    public void reportDuty() {
        System.out.println("Employee reports for duty");
    }
}
class WarehouseSupervisor extends Employee {
    public void monitorStock() {
        System.out.println("Stock monitored by supervisor");
    }
    public void overseeShifts() {
        System.out.println("Shifts overseen by supervisor");
    }
}
class WarehouseManager extends WarehouseSupervisor {
    public void monitorStock() {
        System.out.println("Stock managed at warehouse level");
    }
}
public class Test {
    public static void main(String[] args) {
        WarehouseSupervisor obj = new WarehouseManager();
        obj.reportDuty();           // LINE 1
        obj.overseeShifts();
        obj.monitorStock();        // LINE 2
    }
}
```

Choose the correct option.

Options :

6406534736754. ✖

LINE 1 generates compilation error because method `reportDuty()` cannot be invoked on `obj`.

This code generates the below output followed by runtime error at LINE 2 because there is ambiguity in which `monitorStock()` method is being invoked.

Employee reports for duty

6406534736755. ✖ Shifts overseen by supervisor

This program generates the output:

Employee reports for duty

Shifts overseen by supervisor

6406534736756. ✔ Stock managed at warehouse level

This program generates the output:

Employee reports for duty

Shifts overseen by supervisor

6406534736757. ✖ Stock monitored by supervisor

Question Number : 131 Question Id : 6406531415452 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

Identify the correct statement to fill in the blank at LINE 1, such that the output is:

Login Successful

```
interface AuthStatus {
    void showMessage();
}
class LoginSystem {
    private boolean isValidUser;
    public LoginSystem(boolean isValid) {
        isValidUser = isValid;
    }
    public AuthStatus authenticate() {
        if (isValidUser)
            return new LoginSuccessful();
        return new LoginFailed();
    }
    private class LoginSuccessful implements AuthStatus {
        public void showMessage() {
            System.out.println("Login Successful");
        }
    }
    private class LoginFailed implements AuthStatus {
        public void showMessage() {
            System.out.println("Login Failed");
        }
    }
}
public class Test {
    public static void main(String[] args) {
        LoginSystem user1 = new LoginSystem(false);
        LoginSystem user2 = new LoginSystem(true);
        ----- // LINE 1
    }
}
```

Options :

6406534736758. ✖ user1.showMessage();

6406534736759. ✖ user2.showMessage();

6406534736760. ✔ user2.authenticate().showMessage();

6406534736761. ✖ user1.authenticate().showMessage();

Question Number : 132 Question Id : 6406531415453 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the Java code given below.

```
abstract class Channel {
    public abstract void broadcast();
    public void channelInfo() {
        System.out.println("This is a TV channel");
    }
}
abstract class NewsChannel extends Channel {
    public void broadcast() {
        System.out.println("Broadcasting live news");
    }
    public abstract void exclusiveSegment();
}
class SportsChannel extends NewsChannel {
    public void exclusiveSegment() {
        System.out.println("Showing match highlights");
    }
}
public class Test {
    public static void main(String[] args) {
        Channel obj = new SportsChannel(); // LINE 1
        obj.channelInfo();
        obj.broadcast();
        obj.exclusiveSegment();           // LINE 2
    }
}
```

Choose the correct option:

Options :

The output will be:

This is a TV channel

Broadcasting live news

6406534736762. ✖ Showing match highlights

The code will print:

This is a TV channel

Broadcasting live news

6406534736763. ✖ followed by a runtime error.

Compilation error at LINE 1 because Channel cannot refer to an object of

6406534736764. ✖ type SportsChannel.

LINE 2 generates compilation error because the method `exclusiveSegment()` cannot be invoked on obj 6406534736765. ✓

Sub-Section Number : 3
Sub-Section Id : 640653228429
Question Shuffling Allowed : Yes

Question Number : 133 Question Id : 6406531415445 Question Type : MCQ
Correct Marks : 5

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.util.*;
import java.util.stream.*;
class Student {
    String name;
    int marks;
    int attendance;
    Student(String n, int m, int a) {
        name = n;
        marks = m;
        attendance = a;
    }
    public String toString() {
        return name;
    }
}
public class ResultProcessor {
    public static void main(String[] args) {
        var students = List.of(
            new Student("Alice", 85, 80),
            new Student("Bob", 55, 90),
            new Student("Charlie", 70, 65),
            new Student("Daisy", 62, 78)
        );
        Map<Boolean, List<Student>> result = students.stream()
            .collect(Collectors.partitioningBy(____CODE____)); //LINE 1
        System.out.println(result.get(true));
    }
}
```

Choose the correct option to be filled in place of LINE 1 so that the output is: [Alice, Daisy]

Options :

6406534736729. ✖ s -> s.marks > 60 || s.attendance > 75

6406534736730. ✔ s -> s.marks >= 60 && s.attendance >= 75

6406534736731. ✖ s -> s.attendance >= 75

6406534736732. ✖ s -> s.marks >= 60

Question Number : 134 Question Id : 6406531415454 Question Type : MCQ

Correct Marks : 5

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import java.util.*;
public class MapTest {
    public static void printStudents(Map<String, Integer> m) {
        var map1 = new LinkedHashMap<String, Integer>();
        var map2 = new TreeMap<String, Integer>();
        String[] students = {"Neha", "Arjun", "Kunal"};
        for (String s : students) {
            if (m.containsKey(s)) {
                map1.put(s, m.get(s));
                map2.put(s, m.get(s));
            } else {
                map1.put(s, 0);
                map2.put(s, 0);
            }
        }
        System.out.println(map1);
        System.out.println(map2);
    }
    public static void main(String[] args) {
        var marks = new HashMap<String, Integer>();
        marks.put("Arjun", 76);
        marks.put("Kunal", 93);
        marks.put("Neha", 91);
        marks.put("Divya", 88);
        printStudents(marks);
    }
}
```

What will the output be?

Options :

6406534736766. ✖

```
{Neha=91, Kunal=93, Arjun=76}  
{Arjun=76, Kunal=93, Neha=91}
```

```
        {Neha=91, Arjun=76, Kunal=93}  
6406534736767. ✓ {Arjun=76, Kunal=93, Neha=91}
```

```
        {Neha=91, Arjun=76, Divya=88}  
6406534736768. ✗ {Arjun=76, Neha=91, Divya=88}
```

```
        {Neha=91, Arjun=76, Kunal=93, Divya=88}  
6406534736769. ✗ {Arjun=76, Neha=91, Kunal=93, Divya=88}
```

Sub-Section Number : 4
Sub-Section Id : 640653228430
Question Shuffling Allowed : Yes

Question Number : 135 Question Id : 6406531415448 Question Type : MCQ

Correct Marks : 6

Question Label : Multiple Choice Question

Consider the Java code given below.

```
import javax.swing.*;  
import java.awt.event.*;  
  
public class Test extends JFrame implements ActionListener {  
    JButton b = new JButton("Click");  
    JLabel l = new JLabel("Initial");  
  
    public Test() {  
        add(b, "North"); add(l, "South");  
        setSize(200, 100); setVisible(true);  
    }  
  
    public void actionPerformed(ActionEvent e) {  
        l.setText("Clicked");  
    }  
  
    public static void main(String[] args) {  
        new Test();  
    }  
}
```

What will the output be when the button labelled "Click" is clicked?

Options :

6406534736741. ✖ The label will change to "Clicked" when the button is clicked

6406534736742. ✔ The label remains "Initial"

6406534736743. ✖ The code throws a runtime error

6406534736744. ✖ The button will disappear after clicking

Sub-Section Number :

5

Sub-Section Id :

640653228431

Question Shuffling Allowed :

Yes

Question Number : 136 Question Id : 6406531415455 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below.

```
class Container<T extends Number> {
    private T value1;
    private T value2;
    public Container(T v1, T v2) {
        value1 = v1;
        value2 = v2;
    }
    public T getValue1() {
        return value1;
    }
    public T getValue2() {
        return value2;
    }
    public Container<Integer> addValues(ARG) { // LINE 1
        Container<Integer> newContainer = new Container<>(0, 0);
        newContainer.value1 = this.value1.intValue() + c.getValue1().intValue();
        newContainer.value2 = this.value2.intValue() + c.getValue2().intValue();
        return newContainer;
    }
}

public class Test {
    public static void main(String[] args) {
        Container<Integer> c1 = new Container<>(50, 100);
        Container<Integer> c2 = new Container<>(30, 70);
        Container<Integer> newContainer = c1.addValues(c2);
        System.out.println(newContainer.getValue1() + "," +
                           newContainer.getValue2());
    }
}
```

Choose all the options that can be used in place of ARG in LINE 1 so that the code outputs the sum of two containers.

Options :

6406534736770. ✓ Container<T> c

6406534736771. ✗ Container<Number> c

6406534736772. ✓ Container<?> c

6406534736773. ✗ Container<Object> c

Question Number : 137 Question Id : 6406531415457 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the code given below.

```
class ParkingLot {
    private int spotsAvailable = 2;
    public synchronized void parkCar(String driverName) {
        if (spotsAvailable > 0) {
            System.out.println(driverName + " parked successfully.");
            spotsAvailable--;
        } else {
            System.out.println(driverName + " could not find a parking spot.");
        }
    }
}

class Driver implements Runnable {
    private ParkingLot lot;
    private String name;
    public Driver(ParkingLot l, String n) {
        lot = l;
        name = n;
    }
    public void run() {
        lot.parkCar(name);
    }
}

public class ParkingTest {
    public static void main(String[] args) {
        ParkingLot lot = new ParkingLot();
        Thread d1 = new Thread(new Driver(lot, "X"));
        Thread d2 = new Thread(new Driver(lot, "Y"));
        Thread d3 = new Thread(new Driver(lot, "Z"));
        d1.start();
        d2.start();
        d3.start();
    }
}
```

Which of the following options is/are possible result/s of the above code?

Options :

X parked successfully.

Y parked successfully.

6406534736778. ✓ Z could not find a parking spot.

Z parked successfully.

X parked successfully.

6406534736779. ✓ Y could not find a parking spot.

```
Y parked successfully.
X parked successfully.
6406534736780. ✖ Z parked successfully.

Y could not find a parking spot.
Z parked successfully.
6406534736781. ✖ X could not find a parking spot.
```

Sub-Section Number : 6
Sub-Section Id : 640653228432
Question Shuffling Allowed : Yes

Question Number : 138 Question Id : 6406531415456 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below.

```
class SequencePrinter implements Runnable {
    private boolean stopFlag = false;
    private String[] sequence = {"Red", "Green", "Blue", "Yellow"};
    private int idx = 0;
    public void run() {
        while (!stopFlag) {
            System.out.print(sequence[idx] + " ");
            idx = (idx + 1) % sequence.length;
        }
    }
    public void requestStop() {
        stopFlag = true;
    }
}

public class ColorTest {
    public static void main(String[] args) throws InterruptedException {
        SequencePrinter sp = new SequencePrinter();
        Thread thread = new Thread(sp);
        thread.start();
        sp.requestStop();
    }
}
```

Choose the correct option(s).

Options :

6406534736774. ✖ The program will always output: Red Green Blue Yellow

6406534736775. ✖ The program will always output: Red

The output may be any prefix of the sequence Red Green Blue Yellow Red
6406534736776. ✔ Green..

6406534736777. ✔ It is possible that no output is produced at all.

Question Number : 139 Question Id : 6406531415458 Question Type : MSQ

Correct Marks : 6 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the Java code given below.

```
import javax.swing.*;
import java.awt.event.*;

public class ToggleLabel extends JFrame implements ActionListener {
    JButton b = new JButton("Toggle");
    JLabel l = new JLabel("ON");

    public ToggleLabel() {
        JPanel p = new JPanel();
        p.add(b); p.add(l); add(p);
        b.addActionListener(this);
        setSize(200, 100); setVisible(true);
    }

    public void actionPerformed(ActionEvent e) {
        // CODE SEGMENT
    }

    public static void main(String[] args) {
        new ToggleLabel();
    }
}
```

Choose the correct options to be filled in place of CODE SEGMENT so that it correctly toggles the label between "ON" and "OFF".

Options :

```
if (l.getText().equals("ON"))
    l.setText("OFF");
else
```

6406534736782. ✔ l.setText("ON");

6406534736783. ✖ l.setText("OFF");

```
String s = l.getText();  
if (s.equals("ON"))  
    l.setText("OFF");  
else
```

6406534736784. ✔ l.setText("ON");

```
if (b.getText().equals("ON"))
```

6406534736785. ✖ l.setText("OFF");

Appdev2

Section Id :	64065399548
Section Number :	7
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	29
Number of Questions to be attempted :	29
Section Marks :	100
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	No
Section Maximum Duration :	0
Section Minimum Duration :	0
Section Time In :	Minutes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653228433
Question Shuffling Allowed :	No

Question Number : 140 Question Id : 6406531415459 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MODERN APPLICATION DEVELOPMENT II (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?